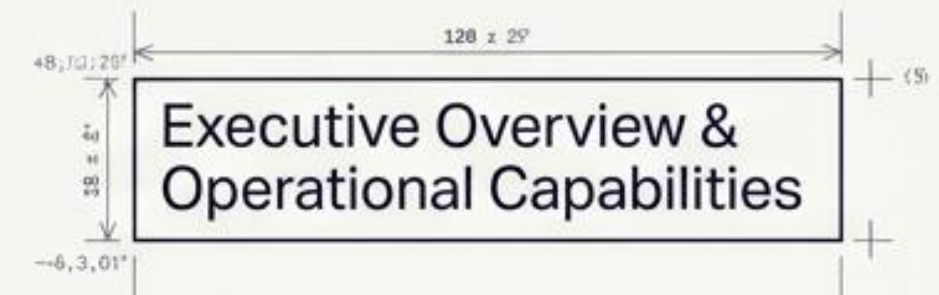
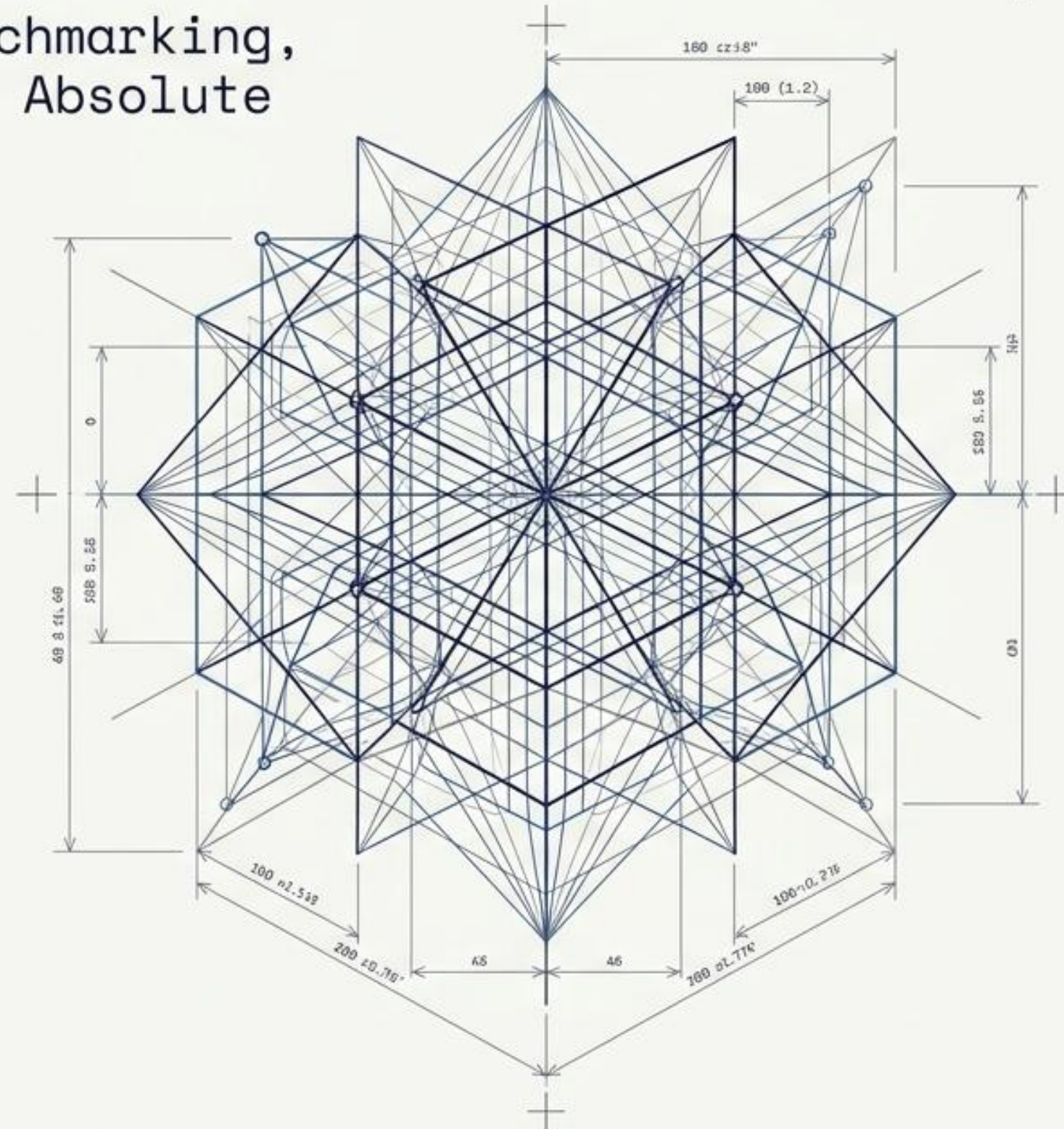
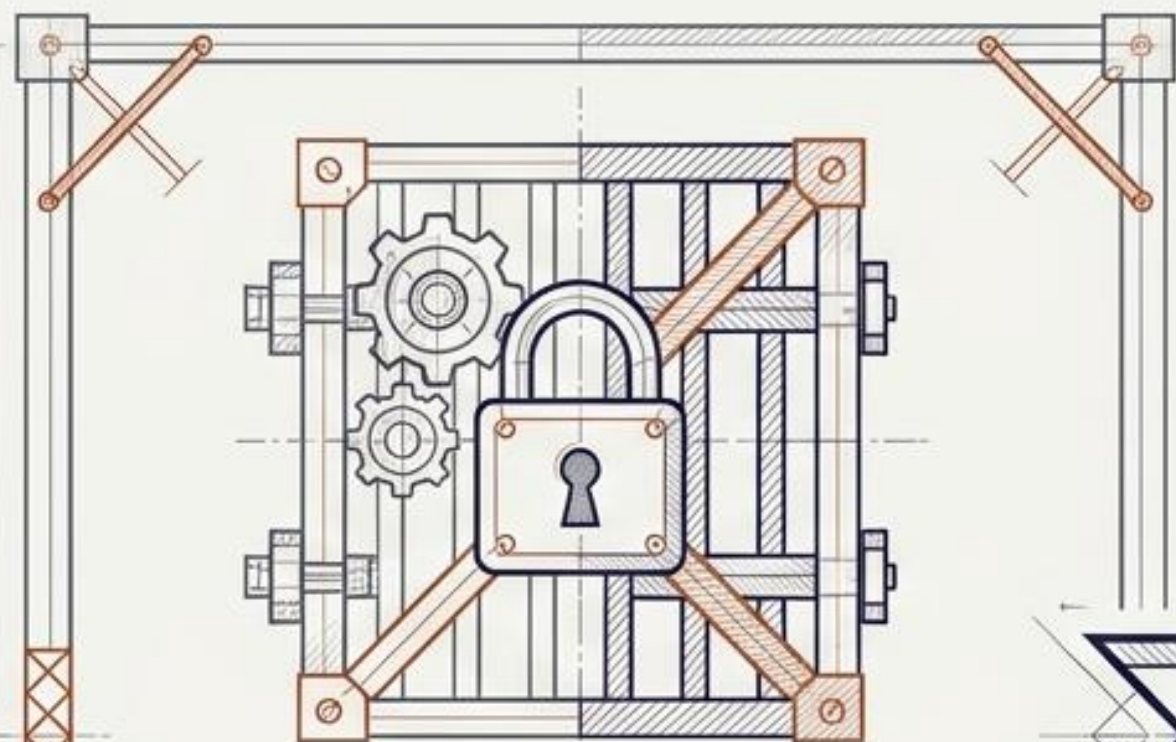


Autobench v3.0: The Precision Compliance Engine

Next-Generation Benchmarking,
Minimal Distortion, Absolute
Control.



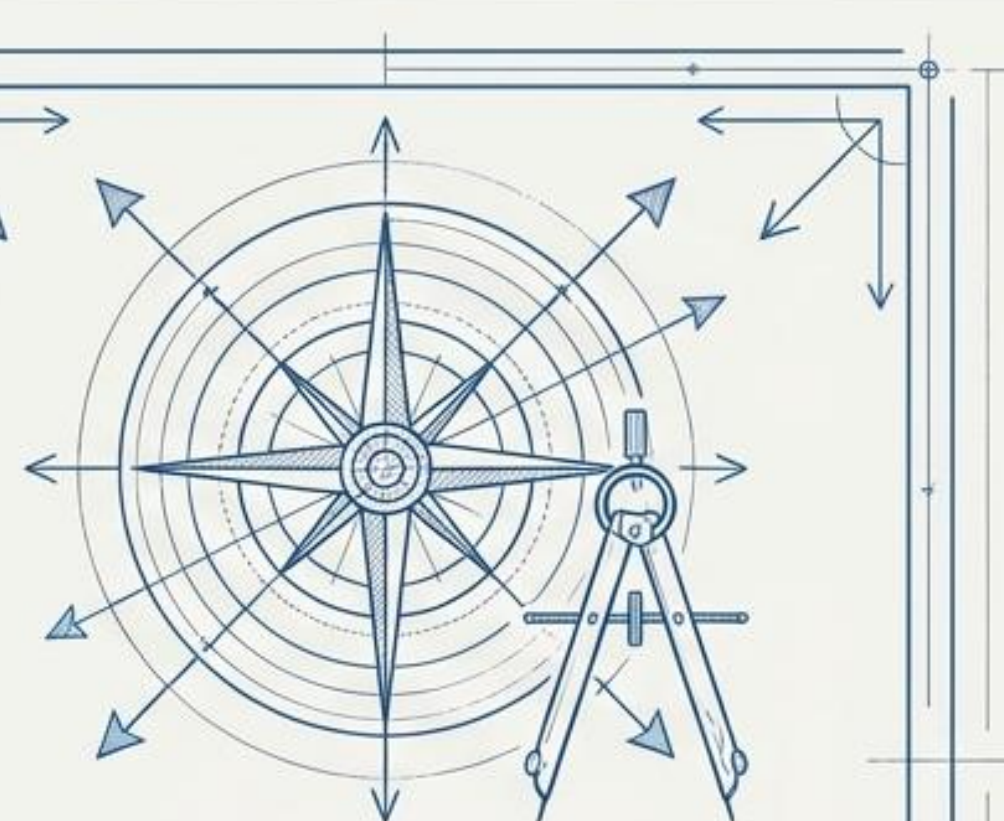
The Paradox of Compliant Benchmarking



The Risk: Control 3 Constraints

- Strict concentration caps (e.g., no peer over 25% share)
- Minimum participant thresholds
- Zero-tolerance regulatory environment
- Manual compliance weighting destroys data fidelity

The Autobench Imperative

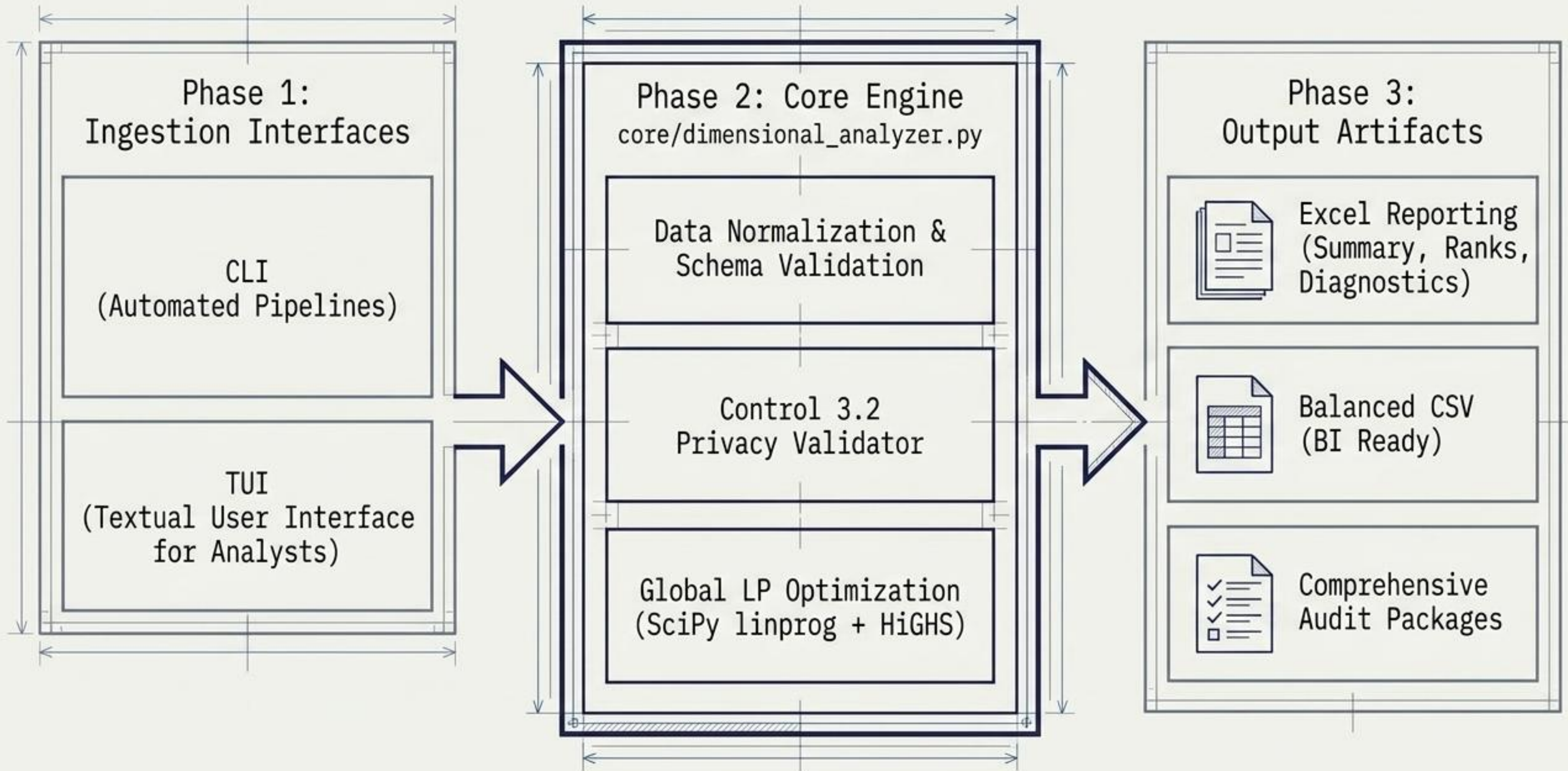


The Goal: Market Reality

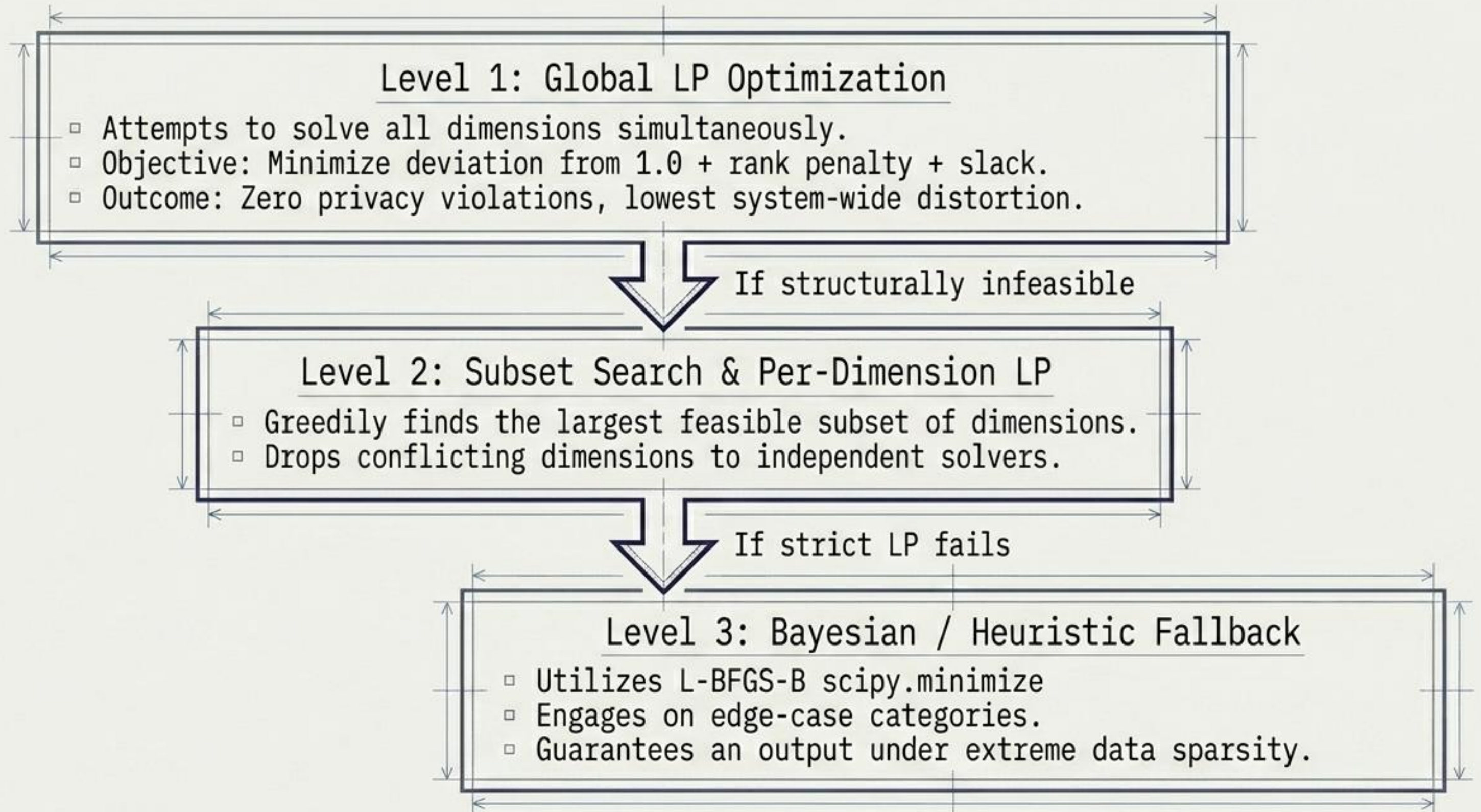
- Maintain exact market positioning
- Preserve rank integrity across dimensions
- Deliver actionable merchant and issuer performance data

The Solution: Autobench v3.0 utilizes advanced Linear Programming (LP) to enforce absolute privacy compliance while mathematically minimizing distortion.

The Autobench Architecture



Algorithmic Resilience: The Solver Fallback



Auto-Applied Control 3.2 Caps

The engine autonomously selects the appropriate regulatory rule based on available peer data—no manual configuration required.

Rule Name	Min Peers	Max Concentration Cap	Trigger Condition
[5/25]	[5]	[25%]	Baseline standard
[6/30]	[6]	[30%]	>= 3 participants at >= 7%
[7/35]	[7]	[35%]	>= 2 at >= 15%, plus >= 1 at >= 8%
[10/40]	[10]	[40%]	>= 2 at >= 20%, plus >= 1 at >= 10%
[4/35]	[4]	[35%]	Merchant benchmarking mode only

Tailoring the Engine: Strategic Presets

Out-of-the-box configurations mapped to specific business intents.



Balanced Default

`balanced_default.yaml`

Use Case: Day-to-day general business analysis.

Posture: Optimal mix of speed, compliance, and accuracy.

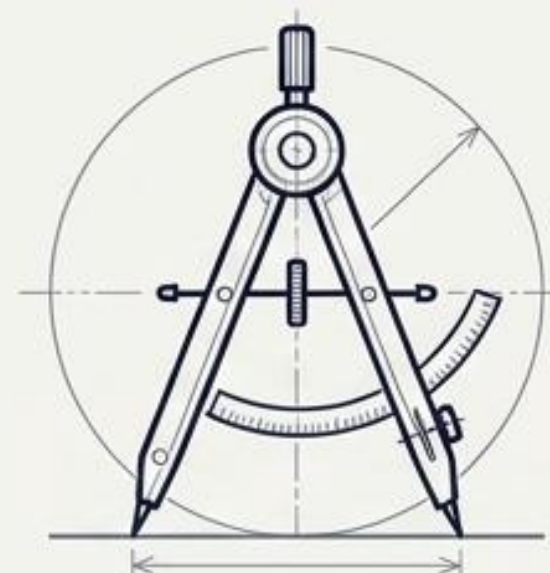


Compliance Strict

`compliance_strict.yaml`

Use Case: Regulatory and audit-first submissions.

Posture: Zero tolerance for threshold deviations; robust audit logs.



Strategic Consistency

`strategic_consistency.yaml`

Use Case: Executive dashboards and KPI tracking.

Posture: Emphasizes consistent global weighting behavior over time.



Research Exploratory

`research_exploratory.yaml`

Use Case: Hard, sparse, or highly skewed datasets.

Posture: Higher flexibility to uncover underlying market trends.

The Analyst Experience: Visual Validation

v3.0 introduces a robust TUI, eliminating pure-CLI friction.

4. Instant Overrides

Toggle advanced panels (Ctrl+A) to inject custom rules instantly.

2. Intuitive Configuration

Dropdown selections for Entity IDs, Metrics, and Dimensions.

The dashboard is titled "ANALYST VALIDATION DASHBOARD - v3.0". It features a top status bar with three sections: "SCHEMA CHECK: PASS", "NULL DETECTION: PASS", and "PEER CONSTRAINTS: VALIDATING..." with a progress indicator. Below this is a "CONFIGURATION" panel with dropdowns for "Entity ID:" (showing "[SELECT ENTITY]"), "Metric:" (showing "[SELECT METRIC]"), and "Dimension:" (showing "[SELECT DIMENSION]"). It also includes a "Date Range:" field with two input boxes and a minus sign, and a "Filter Rules:" text input. To the right is a "DATA PREVIEW" table with columns "ID", "TIMESTAMP", "VALUE", and "STATUS". The table contains 10 rows of data, all with a "VALID" status. At the bottom is a "DIAGNOSTIC LOGS" panel with a close button (X) and a log of four messages. On the far right, a vertical sidebar labeled "ADVANCED OVERRIDES (Ctrl+A)" is partially visible.

ID	TIMESTAMP	VALUE	STATUS
E123	2024-05-20T10:00	150.5	VALID
E122	2024-05-20T10:00	150.5	VALID
E123	2024-05-20T10:00	150.5	VALID
E144	2024-05-20T10:00	150.5	VALID
E155	2024-05-20T10:00	150.5	VALID
E156	2024-05-20T10:00	150.5	VALID
E127	2024-05-20T10:00	150.5	VALID
E138	2024-05-20T10:00	150.5	VALID
E139	2024-05-20T10:00	150.5	VALID

DIAGNOSTIC LOGS

- [10:00:01] INFO: Initializing validation checks...
- [10:00:02] WARN: Potential schema mismatch in file_01.csv, attempting auto-fix...
- [10:00:03] ERROR: Peer constraint violation detected: Entity E456 < min_peers.
- [10:00:04] INFO: Background thread processing metrics... UI responsive.

1. Validation-First Workflow

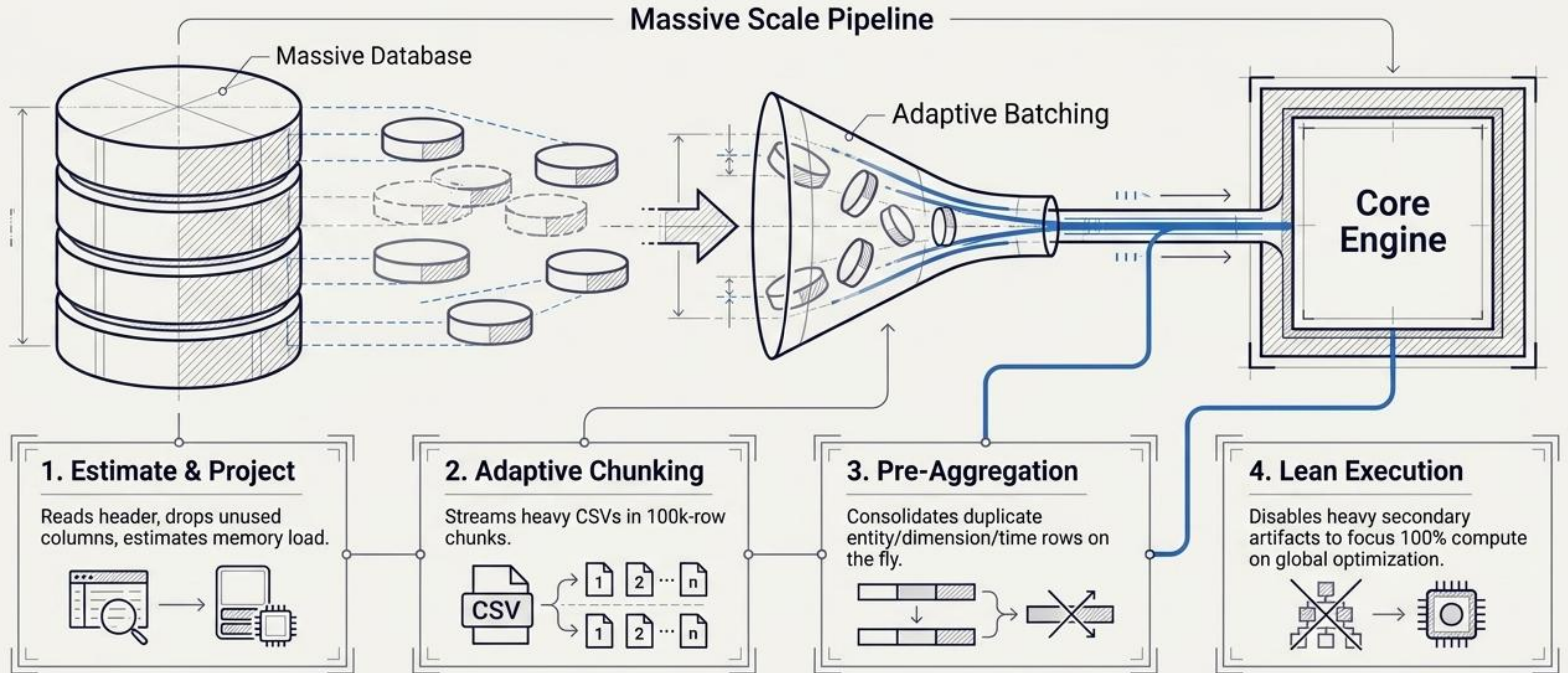
Catches nulls, schema errors, and peer constraints before running heavy optimizations.

3. Real-time Diagnostics

Integrated log handling prevents UI freezes during background thread processing.

Resource Management: Adaptive Batching & Lean Mode

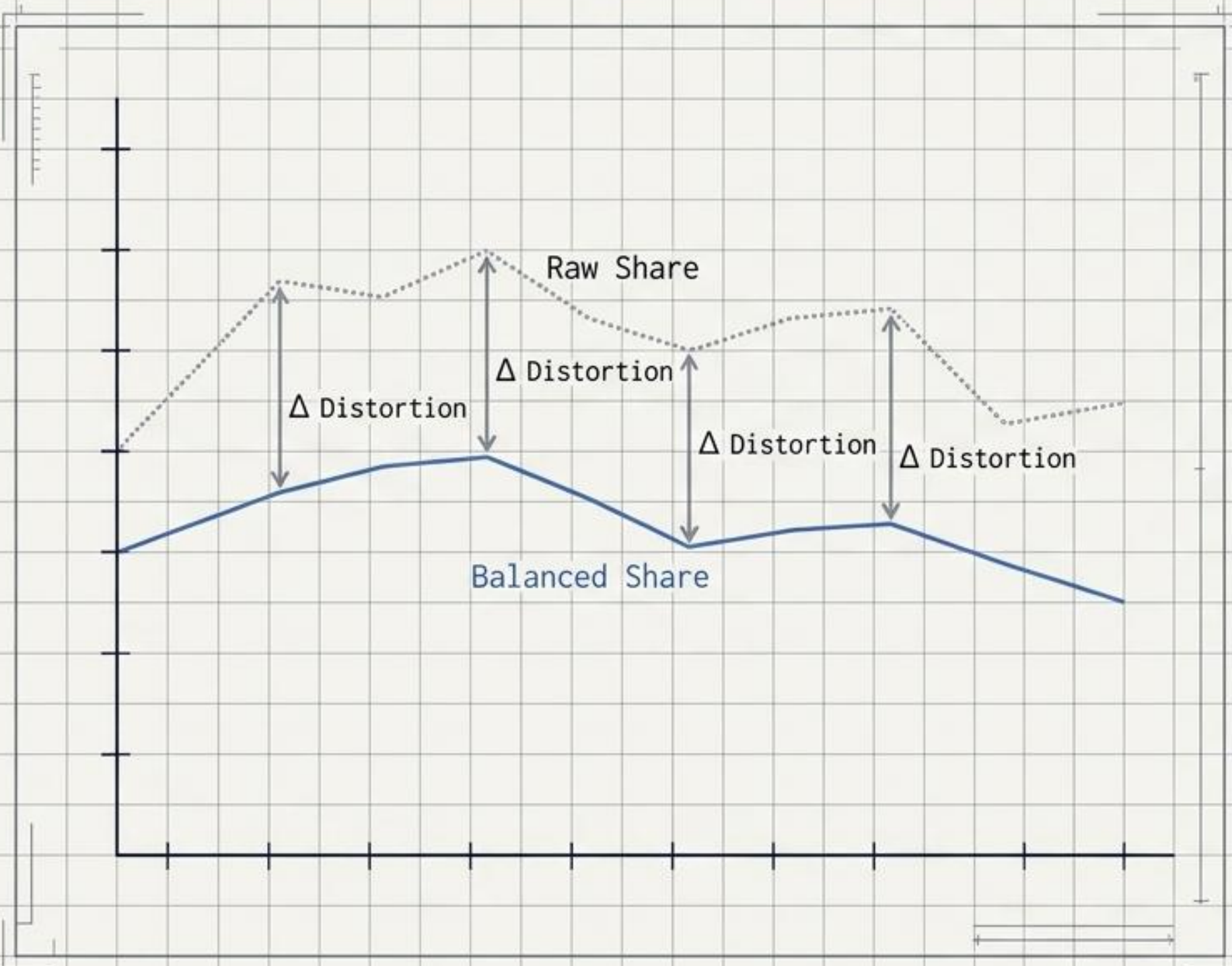
Engineered for memory-limited environments and massive datasets.



Takeaway: Uncompromised privacy cap enforcement with a fraction of the memory footprint.

Unprecedented Visibility into Distortion Impact

What is the true cost of meeting privacy caps?



Distortion Analysis

v3.0 automatically calculates the percentage point delta between raw market truth and compliant reporting.

Rank Change Tracking

A dedicated output sheet tracks exact rank shifts before and after weighting.

Preset Comparison

Run `--compare-presets` to generate a matrix showing which configuration yields the lowest mean distortion for a specific dataset.

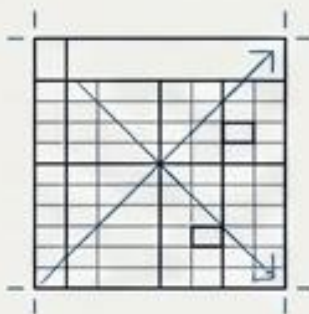
Insight: We don't just enforce rules; we quantify the mathematical impact of those rules on our intelligence.

Embedded Policy Enforcement

The system actively guards against unauthorized data combinations.

Data Scenario	Policy Condition	Status Badge
Fraud/Chargebacks	Must set privacy_basis: clearing_spend	[ENFORCED]
Digital Wallets	Requires explicit Privacy team review	[MANUAL APPROVAL REQUIRED]
Dual-Entity-Axis	Requires explicit Privacy team review	[MANUAL APPROVAL REQUIRED]
Top-Merchant Lists	Strictly prohibited	[HARD BLOCKED]

Actionable Intelligence & Audit Readiness



1. The Excel Report (.xlsx)

- Executive Summary & Metadata.
- Category-level comparisons (Target vs. Best-in-Class).
- Rank Changes & Weight Methods.



2. The Balanced Export (.csv)

- BI-Ready (Tableau, Power BI).
- Enriched with calculated metrics (Distortion_PP, Raw_Share).
- Schema-validated for parity with Excel.

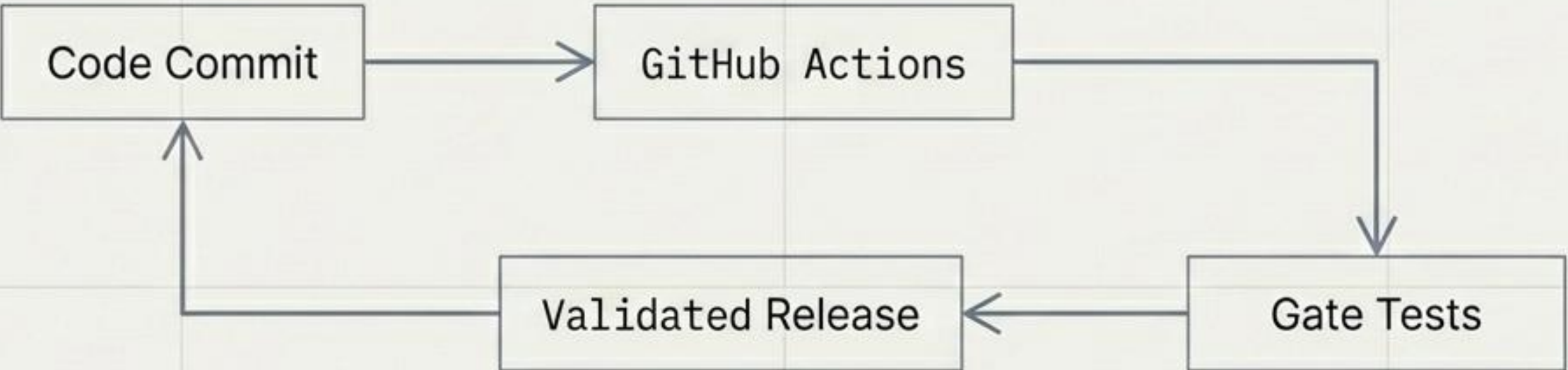


3. The Audit Package (.zip)

- Immutable snapshot of inputs, config, and validation logs.
- Ready for immediate regulatory submission.

Zero Trust, Verified Execution

Our release pipeline ensures absolute mathematical integrity.



Gate Test Suite	Mathematical Parity Validation	Immutable Offline Deployments
17+ representative scenarios running on every commit.	Automated cross-checking to ensure CSV exports match Control 3.2 compliant Excel outputs to an exact 0.01% tolerance threshold.	Secure, checksum-verified offline bundling (offline_packages/) for air-gapped or restricted server environments.

Autobench v3.0: Value Realization

Absolute Compliance

- 100% adherence to Control 3.2 privacy caps.
- Cryptographically verifiable audit trails.

Mathematical Superiority

Industry-leading LP solvers ensure the lowest possible data distortion.

Unlocks insights in sparse datasets that legacy tools abandon.

Operational Agility

TUI and Lean Mode democratize access and scale to massive datasets.

Requires no infrastructure bloat.

Compliant Data is no longer a compromise. It is an exact science.